

Element F

The Nashville Zoo Veterinary Hospital serves to provide animal healthcare, providing essential care to a diverse range of species. In emergency situations where heavy animals require medical attention, the ability to lift and maintain them safely and securely is crucial to the hospital.

Recognizing this need, our project aims to design a heavy animal lift that meets the criteria verified by our stakeholders. We have ensured that our project closely follows the established range of criteria, including cost constraints, loading time, weight-bearing capacity, durability under sustained use, ease of operation, balance, removability, and the mitigation of any potential discomfort or harm to the animals involved.

Our design for the heavy animal lift centers around a sturdy framework engineered from bolted electro-galvanized superstrut channels, reinforced with two-sided angle brackets to provide stability and durability under heavy loads in multiple dimensions. This adaptable structure is designed to work with existing equipment at the Nashville Zoo Veterinary Hospital, incorporating a method for reusing ratchet straps to securely connect the lift to the provided green tarp that is currently used for the Zoo's field operations. We have taken numerous measures to make the design as ergonomic and easily usable as possible by incorporating many adaptable features and automated machine parts.



We chose to take the route of utilizing a galvanized steel such as Superstrut due to the extreme durability and its improbability to bend under large loads. We initially settled on the material known in the industrial world as Unistrut. Unistrut is a versatile umbrella of steel channels, fittings, and accessories designed for constructing support structures in various applications. Its properties include high strength, durability, and corrosion resistance. According to Unistrut Buffalo Supports, “Unistrut is the original metal framing system featuring a unique weldless connection. It eliminates welding and drilling, and is easily adjustable and reusable, for infinite configurations. Over time, the brand has evolved from a simple connection concept to a comprehensive engineered building and support system featuring a robust line of: Channels, Fittings, Fasteners, Hangers, Pipe Clamps, and Accessories.” Since, Unistrut offers compatibility with a wide range of accessories, it allows for customization and adaptability to different project requirements. In other words, Unistrut is valued for its efficiency, reliability, and versatility in many applications.

However, after consulting the brilliant engineer and research assistant Bill Rodriguez at Vanderbilt University, we were introduced to a material called Superstrut, which is essentially Unistrut that is designed for smaller scale applications. He suggested that this material would be ideal for our application due to its cost-effectiveness and durability under loads similar to what our design will experience.

According to Clayton S. Johnson, an engineer who has experience in utilizing both Unistrut and Superstrut, Superstrut is a versatile metal framing system gaining popularity for its flexibility, durability, and customization options across various applications like electrical, mechanical, and industrial support. Its pros include versatility, ease of installation without welding or drilling, and a high load capacity. However, it can be pricier in bulk than its rival, Unistrut, due to higher-quality materials and a wider range of fittings. Additionally, its unique design may limit compatibility with other strut systems, affecting integration with existing structures.

In the context of our design, however, we were not purchasing the Superstrut in bulk, nor did we require it to work with Unistrut-proprietary parts, so we decided that Superstrut was best for our project. We are extremely confident in the durability of the Superstrut not to bend under extreme loads close to the 400 to 1100 pounds specified by the veterinary hospital, and thus believe the framework of our design to be a viable solution to our task.

Another strength of our project was adapted from a suggestion by Bill Rodriguez to help keep the lift balanced and create a point of attachment for the lift to the industrial coffering bottom hook

at the vet hospital. Mr. Rodriguez is experienced in working with heavy automobile engines and often uses a bar engine leveler to remove engines from cars and keep them balanced as needed.



The engine leveler, as shown above, allows the user to adjust the fulcrum of the balance and maintain the balance of a system regardless of the changing center of mass. The photo on the right shows how, after slightly adjusting the fulcrum position, you can maintain the balance with an uneven weight distribution. This capability is crucial for the design, as it completely eradicates the past issue of maintaining the balance of the lift, which is essential for the stakeholders at the hospital. Additionally, the load limit of the leveler is 1500 pounds, which exceeds our maximum criteria of 1100 pounds. This system and its advantages help to make the design in its entirety a viable solution to the given problem of the Nashville Zoo.

Our design group got to spend a day with experienced engineers, as well as the Nashville Zoo Vet Hospital's very own Dr. Stephen Wancea, and we were able to collect valuable input and suggestions. NTC Award winning CCTE Director Jeremy Qualls told us that he likes our framework and was impressed with the engine leveler that we utilized in our prototype. Some additional suggestions by Mr. Qualls include a remote actuator for the engine leveler. Taking this into consideration, we determined that we can actuate the fulcrum-adjuster with a handheld drill (with the appropriate drill bit) in order to maximize efficiency and decrease the time required to balance the animal, as this could be problematic for the doctors.

In considering the risks associated with each sub-part of our design, we have established a primary issue brought to our attention by Mr. Wancea. First of all, due to the immense size of many of the zoo's animals, the straps are almost guaranteed to be in contact with the animals throughout the process, meaning that the material used needs to be safe and comfortable when in constant contact with the animal's skin. Dr. Wancea at the Nashville Zoo explained how they often use pool noodles or wire tubing to safeguard against any harm or irritation caused by straps rubbing against the animals' skin. We decided that taking the route of nylon straps would eliminate the problems related to discomfort from the metal cables, a decision to which Dr. Wancea was in agreement.

In summary, our design surpasses all of the given criteria and incorporates many suggestions by both our stakeholders and experienced professionals in the engineering field. These same professionals and stakeholders have assured us of our design's viability and made sure to point out areas that still need to be tested and improved. Our use of galvanized steel framework and

integration of a balancing mechanism show our careful consideration of the given criteria and the usability of our design for the stakeholders.

Works Cited

Johnson, Clayton S. "Superstrut vs. Unistrut: Battle of the Struts." *Clayton Notes*, 7 Apr. 2023, claytonnotes.com/superstrut-vs-unistrut/.

"What Is Unistrut? - UNISTRUT Buffalo Supports." *Unistrut Buffalo Supports, Longest Running Independent Supplier of Unistrut Metal Framing in the World!*, unistrutstore.com/what-is-unistrut. Accessed 13 May 2024. \